AIRSTREET · REGULATORY REGISTER

AirStreet regulatory and standards register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First register (kit 1.9.1 upgrade); the first Smart Cities register, with air quality rows A1 to A6 for other Smart Cities repos to cite

AirStreet regulatory and standards register

Which standards and regulations bear on AirStreet, where, and what is needed. AirStreet is a research and community prototype that gives indicative air quality levels; it is not a regulatory monitor, and nothing here claims compliance with an air quality standard. This is not legal advice; anything marked "to assess" needs a qualified reviewer before any product claim. Rows A1 to A6 are researched once for the Smart Cities focus area and can be cited by other repos in it. Rows marked FieldNode belong to the shared FieldNode core and are assessed in its repository. Method: STANDARDS section 23.

Table 1. Regulatory and standards register.

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
A1	Directive 2008/50/EC on ambient air quality and cleaner air for Europe [1]	European Union	Limit values (NO2 annual 40 ug/m3) and the siting of sampling points (inlets in general 1.5 to 4 m above ground)	AirStreet's 3.0 m inlet is inside the range (PR-011); PR-004 classifies streets against 40 ug/m3 at the directive's 20 degC reference. The annex's other siting rules (distance from the kerb and from junctions) are to check per site. AirStreet data are indicative and do not replace the directive's fixed measurements	Project lead	Met by design for the inlet; does not apply as a compliance claim
A2	Directive (EU) 2024/2881 on ambient air quality and cleaner air for Europe (recast), published 20 November 2024 [2]	European Union	Recast of A1; annual limit values from 2030 of 20 ug/m3 for NO2 and 10 ug/m3 for PM2.5 [3]	Sets a lower NO2 limit than PR-004 uses; at AirStreet's uncertainty a street could be classified against 20 ug/m3 only outside 13.9 to 26.1 ug/m3 (AST-CAL-001 [C5])	Project lead	To assess (AST-DEC-001, open decision 4)
A3	US National Ambient Air Quality Standards: NO2 annual 53 ppb and 1-hour 100 ppb; PM2.5 annual 9.0 ug/m3 and 24-hour 35 ug/m3 [4]	United States	Ambient standards met by regulatory (federal reference or equivalent) monitors	The first partner candidate is in Texas (AST-DEC-001); AirStreet's data can be compared with these levels as indicative values only	Partner organization	Does not apply as a compliance claim; context for a US partner
A4	US EPA air sensor performance testing protocols and targets: PM2.5 and ozone (February 2021); NO2, CO and SO2 (February 2024) [5]	United States (voluntary)	Base and enhanced testing of sensors beside reference monitors, with performance targets	The route for the TRL 4 collocation and for reporting AirStreet's NO2 and PM2.5 performance in a recognised form	Design lead	Applies as practice at TRL 4; to assess against the protocol text
A5	CEN/TS 17660-1:2021, performance evaluation of air quality sensor systems, part 1: gaseous pollutants in ambient air [6]	European Union (technical specification)	Laboratory and field tests and classes for sensor systems used for indicative measurement	Applies to the NO2 channel if a European partner wants a classed result; the particulate part of the series is to assess	Design lead	To assess (the text has not been read; it must be bought)
A6	Defra, diffusion tubes for ambient	United Kingdom	Exposure of 2 to 4 weeks, 2 to 4 m		Design lead	Applies if option C or the

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
	NO2 monitoring, practical guidance [7]	(guidance, widely used)	height, triplicate tubes, bias adjustment by collocation, about 25 % uncertainty	Option C of AST-OPT-001 and the proposed cross-check of open decision 2		cross-check is chosen
R1	ETSI EN 300 220 and the LoRaWAN EU868 regional parameters (1 % duty cycle per sub-band) [8]	European Union and countries using EU868	Duty cycle and power in 863 to 870 MHz	0.60 % at SF12 (PR-010)	Design lead	Met by design on paper (AST-CAL-001 [B3])
R2	FCC 47 CFR Part 15 and the LoRaWAN US915 regional parameters (400 ms dwell time; DR0 at SF10 carries 11 bytes) [9]	United States and countries using US915	Dwell time and payload per data rate	The 20-byte record goes at SF9 or faster, 247 ms (PR-010)	Design lead	Met by design on paper (AST-CAL-001 [B5]); certified radio module in FieldNode
R3	The Things Network fair use policy (30 s of uplink airtime a day per node; 10 downlinks) [10]	Network operator	Airtime per device	15 min records on TTN, 23.7 s a day at SF9 (PR-008)	Design lead	Met by design on paper (AST-CAL-001 [B8])
R4	Radio Equipment Directive 2014/53/EU and national type approval	European Union; each partner country	Placing radio equipment on the market	AirStreet uses FieldNode's certified LoRa module; prototypes built by a partner are not placed on the market	FieldNode lead	To assess (FieldNode)
E1	IEC 60529, degrees of protection by enclosures (IP code)	International	IP65 and other ratings	Electronics in FieldNode's IP65 enclosure; the pod is open below by design (its inlets), so its electronics rely on the drip lid, mesh and conformal coating	Design lead	Met by choice of parts on paper (FieldNode); pod to verify at TRL 4
E2	Lithium cell safety and transport: IEC 62133-2 and UN 38.3	International	Safety and shipping of the LiFePO4 cell	The cell is in the FieldNode core	FieldNode lead	To assess (FieldNode)
E3	Regulation (EU) 2023/1542 on batteries	European Union	Labelling, removability and end of life of batteries	FieldNode's cell, if units are placed on the EU market	FieldNode lead	To assess (FieldNode); not for a research prototype

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E4	WEEE Directive 2012/19/EU and RoHS Directive 2011/65/EU	European Union	End of life and restricted substances in electrical equipment	The sensors, boards and cables; electrochemical cells go back to their maker for disposal	Design lead	To assess before any supply beyond research prototypes
S1	The pole owner's attachment rules and street works permission	Partner city	Permission, load, inspection and insurance for devices on street furniture	Every installation (PR-018); the owner checks the 3.77 kg and 125 N wind load	Partner organization	Applies; to assess with the first pole owner
S2	National rules for work at height and traffic management at the roadside	Partner country	Platforms, ladders and traffic management for street works	Every installation and pod exchange (PR-018, AST-BLD-001 safety stops)	Partner organization	Applies; to assess with the partner
S3	The NO2 cell maker's safety data sheet (acid electrolyte)	Supplier	Handling, storage and disposal of electrochemical cells	Cells are fitted and replaced on the ground and returned for disposal (PR-016)	Design lead	Applies
D1	General Data Protection Regulation (EU) 2016/679 and similar privacy law	European Union and others	Processing of personal data	AirStreet collects levels, T and RH only: no images, audio or identifiers (PR-014)	Design lead	Does not apply (met by design)
D2	The partner's open data licence and the platform's terms (for example OpenAQ)	Partner and platform	Licence and attribution for published data	Calibrated hourly data with the calibration version (AST-DDR-001 D9)	Partner organization	To assess with the partner

Status: to assess, applies, does not apply, met by design, certification needed later.

Sources

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10. The Things Network forum, "The Things Network Fair Use Policy," https://www.thethingsnetwork.org/forum/t/the-things-network-fair-use-policy/47689, accessed 2026-10-24.